Preprint

SUPPLEMENTARY MATERIAL

Supplementary Material: The Constitutive Law from Exchange Symmetry

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Abstract
The tetrad equation produces two parity-related bounded regions, TetA and TetB [1]. Let the acceleration and the fraction μ of field capacity carried by TetA satisfy a polynomial equation. Three premises determine that equation: the endpoint $\mu(0) = 0$; closure of the partition states under the exchange of the regions; and degree one in each variable, the degree in μ counting the directions in which the coupling runs. A short lemma on degrees makes the relation bilinear, and the premises then force the screening identity $x(1-\mu) = \mu$, that is $\mu(x) = x/(x+1)$. The exchange is derived, not assumed: it is the inversion $x \to 1/x$ about the scale a_0, and parity reversing the coordinate $s = 2\mu - 1$. Removing any premise admits explicit alternatives, among them the framework's own bound-state law, so each is necessary. No coupling polynomial, field equation or equilibrium postulate enters the proof.

Keywords: constitutive law, interpolating function, MOND, AQUAL, exchange symmetry, algebra, proof

Definitions

Let $\tilde{x} = g/g_* \in [0, \infty]$ be the acceleration in units of an arbitrary reference g_*; the natural unit is fixed below. Let $\mu(\tilde{x}) \in [0, 1]$ be the fraction of the total field capacity carried by TetA, so that $1 - \mu$ is the fraction carried by TetB, and let

$$\Omega = \frac{\mu}{1-\mu} \in [0, \infty] \tag{1}$$

be the odds ratio between the two regions.

Premises

(E) Endpoint. $\mu(0) = 0$: at zero acceleration the field sits entirely in TetB.

(X) Exchange closure. The main letter [1] and its supplement [2] show that the identity and parity components of the tetrad equation give TetA and TetB, exchanged by spatial parity P. Parity leaves $|g|$ unchanged; it acts on the partition states, sending the state in which TetA carries a share μ to the state in which TetB carries that share. Since P is a symmetry, the set of partition states realised on $[0, \infty]$ is closed under it: if μ is realised at some acceleration, so is $1 - \mu$. Nothing more about the exchange is assumed.

(A) Degree one. The tetrad equation is polynomial, and we take $\tilde{x}$ and μ to satisfy a real polynomial equation $P(\tilde{x}, \mu) = 0$, P irreducible. We require P to have degree one in μ and degree one in $\tilde{x}$. The degree in μ counts the directions in which the coupling runs. The unbound field couples in one direction: matter sources the field, and the field does not source itself [3]. A bound field sources itself, couples in both directions, and obeys $x(1-\mu)^2 = \mu$, of degree two in μ [3]. The degree in $\tilde{x}$ is the exponent of the driving term of the second route [5].

Lemma

Lemma. *Let K be a field and $P \in K[x, y]$ irreducible, of degree $m \geq 1$ in y and $n \geq 1$ in x, with $P(x, y) = 0$. Then $[K(x, y) : K(x)] = m$ and $[K(x, y) : K(y)] = n$. So $y \in K(x)$ and $x \in K(y)$ if and only if $m = n = 1$, that is, $P = axy + bx + cy + d$ with $ad - bc \neq 0$.*

Proof. P is irreducible in $K[x][y]$ and has positive degree in y, so it is primitive over $K[x]$, and by Gauss's lemma it is irreducible in $K(x)[y]$. It is therefore the minimal polynomial of y over $K(x)$, of degree m; by symmetry $[K(x, y) : K(y)] = n$. Finally, $P = (ay + b)x + (cy + d)$ factors exactly when (a, b) and (c, d) are proportional, that is when $ad - bc = 0$. □

The Unique Law

Theorem. *Under (E), (X) and (A), with $x = g/a_0$ for a unique scale a_0, the partition relation is $x(1-\mu) = \mu$, so that $\mu(x) = x/(x+1)$, and the exchange acts on accelerations as $x \to 1/x$.*

Step 1: bilinear. By (A) and the lemma, $P = a\tilde{x}\mu + b\tilde{x} + c\mu + d$ with $ad - bc \neq 0$. By (E), $P(0, 0) = d = 0$, so $bc \neq 0$, and in odds

$$\Omega(\tilde{x}) = \frac{\alpha\tilde{x}}{\gamma\tilde{x} + \delta}, \tag{2}$$

with $\alpha = -b$, $\gamma = a + b$, $\delta = c$ and $\alpha\delta \neq 0$.

Step 2: the pole. By (E) the state $\mu = 0$ is realised, so by (X) the state $\mu = 1$, $\Omega = \infty$, is realised at some $\tilde{x} \in [0, \infty]$. The only pole of Ω

is $\tilde{x} = -\delta/\gamma$, read as ∞ when $\gamma = 0$. If $\gamma\delta < 0$ the pole is positive and simple, and Ω changes sign across it, contrary to $\Omega \geq 0$. If $\gamma\delta > 0$ there is no pole on $[0, \infty]$, since $\Omega(\infty) = \alpha/\gamma$ is finite, contrary to (X). Hence $\gamma = 0$, $a = -b$, and $\Omega = k\tilde{x}$ with $k = \alpha/\delta > 0$ because $\Omega \geq 0$.

Step 3: the scale. Setting $a_0 = g_*/k$ and $x = g/a_0$ gives $\Omega(x) = x$, and by Eq. (1)

$$\boxed{x\,(1-\mu) = \mu, \qquad \mu(x) = \frac{x}{x+1}.} \tag{3}$$

The map $(x, \mu) \to (1/x,\, 1-\mu)$ carries Eq. (3) to itself, and μ is single-valued in x, so the state exchanged with the one at x sits at $1/x$: the exchange is the inversion $\iota(x) = 1/x$, whose fixed point is $x = 1$, that is $g = a_0$. □

Consequences. The argument identifies a_0 as the fixed point of the exchange without evaluating it. The second endpoint, $\mu \to 1$ as $x \to \infty$, is derived. The complement symmetry $\mu(x) + \mu(1/x) = 1$ is derived. The involution property $\iota \circ \iota = \mathrm{id}$, which parity requires through $P^2 = 1$, is not used and is recovered. At the fixed point $\mu(1) = 1/2$.

Parity form. Let $s = 2\mu - 1 \in [-1, 1]$, so that TetA is $s = +1$ and TetB is $s = -1$. Equation (3) becomes

$$x\,(1-s) = 1 + s, \qquad s(x) = \frac{x-1}{x+1}, \tag{4}$$

and the exchange becomes $(x, s) \to (1/x, -s)$. In this coordinate the exchange is parity itself, reversing s as it reverses the rank-one term Ks of the coupling polynomial $f(K, s) = 1 + sK + K^2$, whose normalisation $\big(f(K,s) - f(K,-1)\big)/2K$ is $\mu = (1+s)/2$ [1].

Necessity of the Premises

Removing any one premise admits alternatives satisfying the other two:

Removed	Admitted alternatives
(E)	$\Omega = (x+b)/(bx+1)$, $0 < b < 1$; here $\mu(0) = b/(1+b)$ and the states fill $[\,b/(1+b),\, 1/(1+b)\,]$
(X)	$\Omega = \alpha x/(\gamma x + \delta)$, $\alpha, \gamma, \delta > 0$; here $\mu(\infty) < 1$
(A)	$x^n(1-\mu)^m = \mu$, $n, m \geq 1$, $(n, m) \neq (1, 1)$, of degree n in x and m in μ; $(n, m) = (1, 2)$ is the bound-state law [3]

Each relation in the (A) row is irreducible (Eisenstein at the prime μ of $\mathbb{R}[\mu]$), has a single root $\mu \in [0, 1]$ for each x, and runs from $\mu = 0$ to $\mu = 1$. Without (A) the exchange carries no content: any continuous, strictly increasing μ with $\mu(0) = 0$ and $\mu \to 1$ satisfies (E) and (X).

Relation to the Other Routes

The main letter derives the same law by two further routes: graded rank counting of the tetrad equation, with parity interpolation and rational normalisation [4]; and the AQUAL field equation with single-channel detailed balance [5]. The proof above uses neither. The three routes meet in one relation of degree one in each variable. In the first it is the map $s(x)$: of its three boundary conditions, $s(0) = -1$ is (E), $s(1) = 0$ is the choice of unit and $s(\infty) = +1$ is derived, and its fractional linear form is (A) through the lemma. In the second it is the balance $x^n(1-\epsilon) = \epsilon/K$, of degree n in x and one in ϵ, with $n = 1$ fixed there by deep-field scaling.

Summary

For a coupling of degree one in each variable, closure of the partition states under the exchange of TetA and TetB forces the screening identity $x(1-\mu) = \mu$, whose solution is the constitutive law $\mu(x) = x/(x+1)$; each premise is necessary.

This function was identified empirically by Milgrom [6] and selected by fitting by Famaey and Binney [7]. The present derivation obtains it from the exchange symmetry of the two parity-related roots of the equation that defines the gravitational field.

References

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